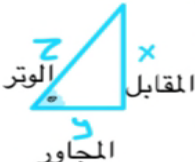


Calculus B

Review

Trigonometric functions

$$\sin \theta = \frac{x}{z} \quad \tan \theta = \frac{x}{y}$$

$$\cos \theta = \frac{y}{z} \quad \tan \theta = \frac{\sin \theta}{\cos \theta}$$


Trigonometric identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sec^2 \theta = 1 + \tan^2 \theta$$

$$\csc^2 \theta = 1 + \cot^2 \theta$$

Double angle

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = 1 - \sin^2 \theta$$

Square angle

$$\sin^2 \theta = \frac{1}{2} (1 - \cos 2\theta)$$

$$\cos^2 \theta = \frac{1}{2} (1 + \cos 2\theta)$$

Sum identities

$$\sin A \cos B = \frac{1}{2} [\sin(A - B) + \sin(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

Special angles

	$\sin \theta$	$\cos \theta$	$\tan \theta$
0	0	1	0
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	1	0	-

Hyperbolic functions

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\operatorname{sech}^2 x = 1 - \tanh^2 x$$

$$\sinh x + \cosh x = e^x$$

Law of Natural logarithm (Ln)

$$\ln(xy) = \ln x + \ln y$$

$$\ln\left(\frac{x}{y}\right) = \ln x - \ln y$$

$$\ln(x^a) = a \ln x$$

Squeeze or sandwich Th. :

If $g(x) \leq f(x) \leq h(x)$
and also suppose that

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L.$$

Then $\lim_{x \rightarrow a} f(x) = L.$

Fundamental theorem of calculus [FTC]

If f is a continuous function on $[a, b]$, then the integral function g defined by

$$g(x) = \int_a^x f(s) ds$$

is continuous on $[a, b]$, differentiable on (a, b) , and $g'(x) = f(x)$.

Properties of limits

- $\lim_{x \rightarrow a} c = c$
- $\lim_{x \rightarrow a} x = a$
- $\lim_{x \rightarrow a} [c f(x)] = c \lim_{x \rightarrow a} f(x)$
- $\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$
- $\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$
- $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \quad \lim_{x \rightarrow a} g(x) \neq 0$

L'Hopital's Rule

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$, then $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$

Derivative Product rule

$$\frac{d}{dx} [f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

Derivative Quotient rule

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

Derivatives of inverse

$\frac{d}{dx} [\sin^{-1} x] = \frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx} [\sec^{-1} x] = \frac{1}{ x \sqrt{x^2-1}}$
$\frac{d}{dx} [\cos^{-1} x] = -\frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx} [\csc^{-1} x] = -\frac{1}{ x \sqrt{x^2-1}}$
$\frac{d}{dx} [\tan^{-1} x] = \frac{1}{1+x^2}$	$\frac{d}{dx} [\cot^{-1} x] = -\frac{1}{1+x^2}$

General derivatives rules

$\frac{d}{dx}(x) = 1$	$\frac{d}{dx}[af(x)] = a \frac{d}{dx}[f(x)]$
$\frac{d}{dx}(ax) = a$	$\frac{d}{dx}(ax^n) = nax^{n-1}$
$\frac{d}{dx}(c) = 0$	$\frac{d}{dx}[f(x)]^n = n[f(x)]^{n-1}f'(x)$

d/dx of trigonometric function

$\frac{d}{dx} [\sin(x)] = \cos x$	$\frac{d}{dx} [\sec(x)] = \sec x \tan x$
$\frac{d}{dx} [\cos(x)] = -\sin x$	$\frac{d}{dx} [\csc(x)] = -\csc x \cot x$
$\frac{d}{dx} [\tan(x)] = \sec^2 x$	$\frac{d}{dx} [\cot(x)] = -\csc^2 x$

d/dx of exponential & logarithmic function

$\frac{d}{dx} [e^x] = e^x$	$\frac{d}{dx} [a^x] = a^x \ln a$
$\frac{d}{dx} [\ln x] = \frac{1}{x}$	$\frac{d}{dx} [\ln x] = \frac{1}{x}, x > 0$
$\frac{d}{dx} [\log_a x] = \frac{1}{x \ln a}$	$\frac{d}{dx} [\ln f(x)] = \frac{f'(x)}{f(x)}$
$\frac{d}{dx} [e^{f(x)}] = f'(x)e^{f(x)}$	$\frac{d}{dx} [a^{f(x)}] = a^{f(x)} \ln a f'(x)$

Chain rule

$$\frac{d}{dx} [(f(x))^n] = n(f(x))^{n-1} \cdot f'(x)$$

$$\frac{d}{dx} [f(g(x))] = f'(g(x))g'(x)$$

Some Integration formulas

$$\int 1 dx = x + C$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int a dx = ax + C$$

$$\int e^x dx = e^x + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

$$\int a^x dx = \frac{a^x}{\ln a} + C, a > 0, a \neq 1$$

$$\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{u}{a}$$

$$\int \frac{du}{u^2 - a^2} = \frac{1}{2a} \ln \left(\frac{u-a}{u+a} \right) = -\frac{1}{a} \coth^{-1} \frac{u}{a} \quad u^2 > a^2$$

$$\int \frac{du}{a^2 - u^2} = \frac{1}{2a} \ln \left(\frac{a+u}{a-u} \right) = \frac{1}{a} \tanh^{-1} \frac{u}{a} \quad u^2 < a^2$$

$$\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \frac{u}{a}$$

Trigonometric substitutions

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta, -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta, 0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2}$	$\sec^2 \theta - 1 = \tan^2 \theta$

Some basic limit :

- $\lim_{n \rightarrow \infty} \frac{\ln(n)}{n} = 0$
- $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$
- $\lim_{n \rightarrow \infty} (x)^{\frac{1}{n}} = 1, x > 0$
- $\lim_{n \rightarrow \infty} (x)^n = \begin{cases} 0, & -1 < x < 1 \\ 1, & x = 1 \\ \infty, & x > 1 \\ DNE, & x \leq -1 \end{cases}$
- $\lim_{n \rightarrow \infty} \frac{(x)^n}{n!} = 0, \text{ for any } x$
- $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x, \text{ for any } x$
- $\lim_{n \rightarrow \infty} \frac{\sin(\frac{k}{n})}{(\frac{k}{n})} = 1, \text{ where } k \in \mathbb{R} \text{ and } (\frac{k}{n}) \rightarrow 0$
- $\lim_{n \rightarrow \infty} \frac{\tan(\frac{k}{n})}{(\frac{k}{n})} = 1$
- $\lim_{n \rightarrow \infty} \frac{1 - \cos(\frac{k}{n})}{(\frac{k}{n})} = 0$

Calculus B

First.

L'Hopital Rule

$$\text{If } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0} \text{ or } \frac{\pm \infty}{\pm \infty}$$

$$\text{then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

7.1 : integration by part

$$\int u dv = uv - \int v du$$

$$u = \quad \quad \quad dv =$$

$$du = \leftarrow \rightarrow v =$$

7.2 : Trigonometric integrals

$$\int \sin^m x \cos^n x dx$$

إذا m أو n فردي $\leftarrow$ نترك العنصر الفردي لا نغير
إذا كلهم زوجي نضربهم كلهم

$$\sin^2 x = \frac{1}{2}(1 - \cos(2x)) \quad \cos^2 x = \frac{1}{2}(1 + \cos(2x))$$

7.2: Trigonometric integrals

$$\int \tan^n x \sec^m x dx$$

$$\int \cot^n x \csc^m x dx$$

If n is odd (tan or cot):

$$\tan^2 x = \sec^2 x - 1, \quad \cot^2 x = \csc^2 x - 1$$

If m is even (sec or csc):

$$\sec^2 x = \tan^2 x + 1, \quad \csc^2 x = \cot^2 x + 1$$

If n is even and m is odd:

I.B.P

7.3: Trigonometric substitution

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta, \quad 0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2}$	$\sec^2 \theta - 1 = \tan^2 \theta$

7.4 : partial fractions

$\frac{f(x)}{g(x)}$
 $\deg f(x) > \deg g(x)$

- long div
- factorize $g(x)$

المقام لا يحلل

 $\left(\frac{Ax+B}{\text{المقام}} + \dots + \frac{\dots}{(\text{المقام})^n} \right)$

$\frac{f(x)}{g(x)}$
 $b^2 - 4a < 0$

إذا المقام لم يفرس
مضرب لا يحلل

$\frac{f(x)}{g(x)}$
 $\deg f(x) < \deg g(x)$

- factorize $g(x)$

المقام يتحلل

 $\left(\frac{A}{Ax+B} + \frac{B}{(Ax+b)^2} + \dots \right)$

7.8 : Improper integrals

Type 1

 $\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$
 $\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$
 $\int_{-\infty}^{\infty} f(x) dx = \lim_{t \rightarrow -\infty} \int_t^0 f(x) dx + \lim_{t \rightarrow \infty} \int_0^t f(x) dx$

Type 2

f has ∞ discont on $[a, b]$

- convergent if the limit exists
- divergent otherwise

Karl weiestrass

$$t = \tan\left(\frac{x}{2}\right) \quad dx = \frac{2 dt}{1+t^2}$$

$$\sin x = \frac{2t}{1+t^2} \quad \cos x = \frac{1-t^2}{1+t^2}$$

11.1 : sequences

- if $\lim_{n \rightarrow \infty} |a_n| = 0$ then $\lim_{n \rightarrow \infty} a_n = 0$
- $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} a_{n-1} = L$
- Seq is decreasing $\rightarrow a_n > a_{n+1} \forall n$, fixed
- Seq is increasing $\rightarrow a_n < a_{n+1} \forall n$, fixed
- Seq is monotonic + bounded $\rightarrow$ conv